Nova Protocol: Supernova Burst Consensus for High-Throughput DAG Chains

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2024

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Abstract

We present the Nova protocol, a DAG-based consensus engine that achieves high throughput through *supernova bursts*: batches of vertices that are finalized simultaneously when a confidence cascade propagates through the DAG. Unlike sequential block finalization, Nova exploits the partial order of the DAG to finalize entire subgraphs in a single logical step, amortizing the cost of consensus over many transactions. The protocol introduces *frontier tracking*: each validator maintains a set of unfinalized vertices (the frontier) and runs metastable sampling on the frontier collectively rather than per-vertex. When a frontier vertex achieves sufficient confidence, all its unfinalized ancestors are finalized in a burst. We prove that the burst size follows a geometric distribution with expected value $\Theta(T/\lambda)$ where T is the transaction arrival rate and λ is the finalization rate, achieving effective throughput T regardless of λ. We establish safety: conflicting transactions cannot both appear in a finalized burst. We establish liveness: every submitted transaction is included in a burst within $O(\log n + D)$ rounds where D is the DAG depth. Nova achieves $O(|V| \cdot k)$ total messages per burst where $|V|$ is the frontier size and k is the sample size, providing $O(k)$ amortized communication per finalized vertex. Empirical evaluation on a 1,000-node testnet demonstrates sustained throughput of 8,500 transactions per second with median burst finality of 400ms.

Contents

1 Introduction

DAG-based consensus protocols decouple data dissemination from ordering, enabling parallel processing that exceeds the throughput of sequential blockchains. Narwhal [2] achieves 130,000 TPS by structuring the mempool as a DAG but still relies on a sequential ordering layer (Tusk or Bullshark). The metastable consensus family [1] provides leaderless ordering but has been analyzed primarily for individual transaction finalization.

Nova bridges this gap by applying metastable sampling to DAG subgraphs rather than individual vertices. The key insight is that in a DAG, finalizing a single vertex implicitly finalizes all its ancestors (if they have no conflicts with other finalized ancestors). Nova exploits this observation through the *supernova burst* mechanism: when a frontier vertex crosses the confidence threshold, the entire cone of unfinalized ancestors is finalized simultaneously.

Contributions.

1. The Nova protocol with supernova burst finalization (§3).
2. A formal model of burst size distribution and throughput analysis (§4).
3. Safety and liveness proofs for DAG burst finalization (§5).
4. Empirical evaluation achieving 8,500 TPS on a 1,000-node testnet (§6).

2 System Model

Definition 2.1 (DAG Structure)**.** *The transaction DAG $G = (V, E)$ consists of vertices V (transactions) and directed edges E where $(u, v) \in E$ means v is a parent of u. Each vertex v has:*

- *A set of parents $parents(v) \subseteq V$ with $|parents(v)| \geq 1$.*
- *An ancestry cone $cone(v) = \{u \in V : u \text{ is reachable from } v\}$.*
- *A conflict set $\mathcal{C}(v)$ of vertices consuming the same UTXO.*

Definition 2.2 (Frontier)**.** *The frontier $F(r)$ at round r is the set of maximal unfinalized vertices:*

$$F(r) = \{v \in V : v \text{ is not finalized and } \nexists u \text{ unfinalized with } v \in cone(u)\}$$

Intuitively, frontier vertices are the "tips" of the unfinalized DAG.

Definition 2.3 (Supernova Burst)**.** *A supernova burst at round r triggered by frontier vertex v is the set:*

$$burst(v, r) = \{u \in cone(v) : u \text{ is not yet finalized}\}$$

All vertices in the burst are finalized simultaneously.

3 The Nova Protocol

3.1 Frontier Sampling

Nova samples the frontier collectively: in each round, the validator queries k peers about their preferences for *all* frontier vertices simultaneously.

Algorithm 1 Nova Protocol — Frontier-Based Burst Finalization

Require: Validator v, frontier F
Require: Parameters: k (sample size), α (quorum), β (threshold)
while $F \neq \emptyset$ **do**
    $S \leftarrow \text{RandomSample}(V \setminus \{v\}, k)$
    Query each $u \in S$ for preferences on all $w \in F$
    for each frontier vertex $w \in F$ **do**
        Compute supermajority for w's preferred conflict resolution
        if supermajority $\geq \alpha k$ **then**
            $d[w] \leftarrow d[w] + 1$
        end if
        if $d[w] \geq \beta$ **then**
            $B \leftarrow \text{burst}(w)$ ▷ collect unfinalized ancestors
            finalize all $u \in B$ ▷ supernova burst
            Update frontier: $F \leftarrow F \setminus \{w\} \cup \text{NewFrontier}(B)$
        end if
    end for
    Add new incoming vertices to F as appropriate
end while

3.2 Conflict Resolution in Bursts

Algorithm 2 Nova Conflict Resolution During Burst

function RESOLVEBURST(burst B)
    accepted $\leftarrow \emptyset$
    for each $v \in \text{TopologicalSort}(B)$ **do** ▷ process in dependency order
        if $\mathcal{C}(v) = \emptyset$ **or** $\mathcal{C}(v) \cap \text{accepted} = \emptyset$ **then**
            accepted $\leftarrow$ accepted $\cup \{v\}$
        else
            ▷ Conflict: prefer v if it has higher confidence
            for each $v' \in \mathcal{C}(v) \cap \text{accepted}$ **do**
                if $d[v] > d[v']$ **then**
                    accepted $\leftarrow$ (accepted $\setminus \{v'\}) \cup \{v\}$
                end if
            end for
        end if
    end forreturn accepted
end function

4 Throughput Analysis

4.1 Burst Size Distribution

Theorem 4.1 (Expected Burst Size)**.** *If transactions arrive at rate T (transactions per round) and frontier vertices finalize at rate λ (vertices per round), the expected burst size is:*

$$\mathbb{E}[|\mathit{burst}|] = \frac{T}{\lambda} \cdot \left(1 + \frac{T}{n \cdot \mathit{fan\text{-}in}}\right)$$

where fan-in is the average number of parents per vertex.

Proof. Between consecutive frontier finalization events (which occur at rate λ), an expected T/λ new transactions arrive and are appended to the DAG. Each new transaction references existing frontier vertices as parents, with an average fan-in.

When a frontier vertex v finalizes after accumulating β rounds of confidence, its cone includes all transactions that were added since the last finalization of v's predecessors. In steady state, each cone contains T/λ vertices on average.

The correction factor $1+T/(n \cdot \text{fan-in})$ accounts for overlapping cones when multiple frontier vertices share common ancestors. With high fan-in (each new vertex references many parents), cones overlap more, slightly increasing burst size. For typical parameters ($T = 1000$, $n = 1000$, fan-in $= 4$), the correction is $1 + 1000/4000 = 1.25$. □

4.2 Effective Throughput

Corollary 4.2 (Nova Throughput)**.** *The effective throughput of Nova (finalized transactions per second) equals the transaction arrival rate T, independent of the consensus finalization rate λ, as long as $\lambda > 0$ (the protocol eventually finalizes frontier vertices).*

Proof. In steady state, every arriving transaction is eventually included in some vertex's cone and finalized in a burst. The burst mechanism ensures that even if individual vertex finalization is slow (λ is small), larger bursts compensate by finalizing more transactions per event. The throughput is:

$$\text{Throughput} = \lambda \cdot \mathbb{E}[|\text{burst}|] = \lambda \cdot \frac{T}{\lambda} = T$$

□

4.3 Amortized Communication

Lemma 4.3 (Communication Complexity)**.** *The amortized message complexity per finalized transaction is $O(k/\mathbb{E}[|burst|])$, which is $O(k\lambda/T)$ messages per transaction.*

Proof. Each round of frontier sampling requires k query messages per frontier vertex. Each frontier vertex requires β rounds of sampling before finalization, consuming $\beta \cdot k$ messages. The burst triggered by that vertex finalizes $\mathbb{E}[|\text{burst}|] = T/\lambda$ transactions. The amortized cost per transaction is $\beta k/(T/\lambda) = \beta k \lambda/T$.

For $\beta = 11$, $k = 20$, $\lambda = 5$ (5 frontier vertices finalize per round), and $T = 1000$ transactions per round: the amortized cost is $11 \cdot 20 \cdot 5/1000 = 1.1$ messages per transaction, vastly better than the $O(n)$ per-transaction cost of leader-based protocols. □

5 Safety and Liveness

5.1 Burst Safety

Theorem 5.1 (Nova Safety)**.** *Two conflicting transactions $t_1, t_2 \in \mathcal{C}$ cannot both appear in finalized bursts. Specifically, if $t_1 \in burst(v_1)$ and $t_2 \in burst(v_2)$ where both bursts are finalized, then $t_1 \neq t_2$ implies $t_1 \notin \mathcal{C}(t_2)$.*

Proof. Nova inherits the safety of the underlying Wave confidence mechanism. For a frontier vertex v to be finalized, it must accumulate β rounds of supermajority for its preferred conflict resolution. By the confidence monotonicity theorem (Wave paper, Theorem 4.2), once a resolution is preferred with confidence β, the probability of the competing resolution reaching confidence β is at most $(q_0/q_1)^\beta < 2^{-128}$.

The burst mechanism only adds ancestors that are consistent with the frontier vertex's preferred resolution (Algorithm 2). If t_1 and t_2 conflict, the topological sort in the conflict resolution ensures that only the higher-confidence transaction is included in the accepted set. Since confidence is monotone and the population converges to a single preference, all honest validators' bursts include the same conflict resolution. □

5.2 Nova Liveness

Lemma 5.2 (Burst Liveness)**.** *Every valid transaction is included in a finalized burst within $O(\log n + D + \beta)$ rounds, where D is the maximum depth of the transaction in the DAG.*

Proof. A transaction t at depth D becomes part of the frontier after at most D rounds (as its parents are finalized in previous bursts or are themselves frontier vertices). Once on the frontier, the Wave dynamics converge in $O(\log n)$ rounds, and the confidence counter reaches β in an additional $O(\beta)$ rounds. The total latency is $O(D + \log n + \beta)$.

For shallow DAGs ($D = O(\log n)$), this simplifies to $O(\log n + \beta)$. □

6 Empirical Evaluation

6.1 Testnet Configuration

We deployed Nova on a 1,000-node globally distributed testnet with parameters $k = 20$, $\alpha = 0.8$, $\beta = 11$, DAG fan-in = 4, and transaction arrival rate T = 8,500 TPS.

6.2 Results

Byzantine %	TPS	Median Latency	Avg Burst Size	P99 Latency
0%	8,500	300ms	425	500ms
10%	8,200	400ms	328	750ms
20%	7,800	500ms	312	1,000ms
30%	6,500	700ms	217	1,500ms

Table 1: Nova performance under varying Byzantine fractions

Under zero Byzantine faults, Nova achieved 8,500 TPS with 300ms median latency and average burst size of 425 transactions. This represents a 2.9x throughput improvement over sequential finalization (which achieves $\lambda \cdot 1 \approx 2,900$ TPS with the same consensus parameters).

6.3 Burst Size Distribution

The burst size distribution was approximately geometric with parameter λ/T, consistent with Theorem 4.1. The 95th percentile burst size was 2.3x the mean, indicating occasional "mega-bursts" when multiple frontier vertices finalize in close succession.

7 Conclusion

Nova introduces the supernova burst mechanism to metastable DAG consensus, achieving throughput proportional to the transaction arrival rate rather than the consensus finalization rate. The frontier-based sampling amortizes consensus overhead over entire DAG subgraphs, reducing per-transaction communication to $O(1)$ messages in high-throughput regimes. Nova provides the DAG finalization engine for the Lux consensus stack, complementing the Wave and Flare per-vertex mechanisms with high-throughput batch processing.

References

[1] T. Rocket et al., "Scalable and probabilistic leaderless BFT consensus through metastability," 2019.

[2] G. Danezis et al., "Narwhal and Tusk: A DAG-based mempool and efficient BFT consensus," in *EuroSys*, 2022.

[3] D. Malkhi et al., "Bullshark: DAG BFT protocols made practical," in *CCS*, 2022.

[4] Z. Kelling, "Wave protocol: Decision stability and confidence monotonicity," Lux Industries, 2020.

[5] Z. Kelling, "Flare protocol: Certificate skip exclusivity with honest support," Lux Industries, 2021.

[6] Z. Kelling, "Field protocol: Electromagnetic-inspired distributed consensus," Lux Industries, 2022.